A Survey on Neural Radiance Fields and Gaussian Splatting in Satellite Remote Sensing

Presenters

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13 April 2026

Importance of 3D Satellite Mapping

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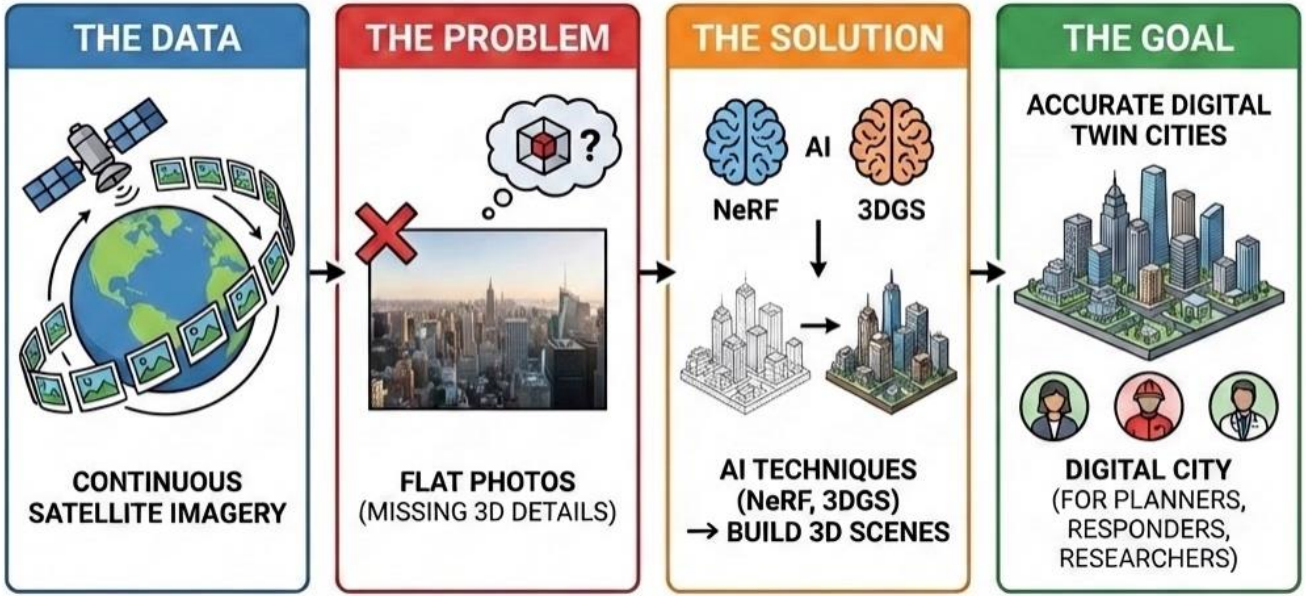
Satellites take thousands of flat photos of our cities and landscapes every day.

But what if we could use those photos to build detailed 3D models of the entire Earth?



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Traditional vs AI-Powered Methods

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	Traditional Mapping	AI-Powered Mapping
Shadows & Reflections	Struggles Heavily	Smart Modeling
Building Edges & Distortions	Blurry Distorted	Razor-Sharp
Textureless Surfaces	Frequently Fails	Seamless Geometery
Seasonal & Weather Changes	Causes Errors	Season Adaptive
Moving (Transient) Objects	Creates “Ghost”	Automatically Removes
Performance & Speed	Slow and Heavy	Real-time (3DGS)

Visual Comparison of Traditional vs AI-Powered Mapping

Traditional Mapping

AI-Powered Mapping (NeRF & 3DGS)

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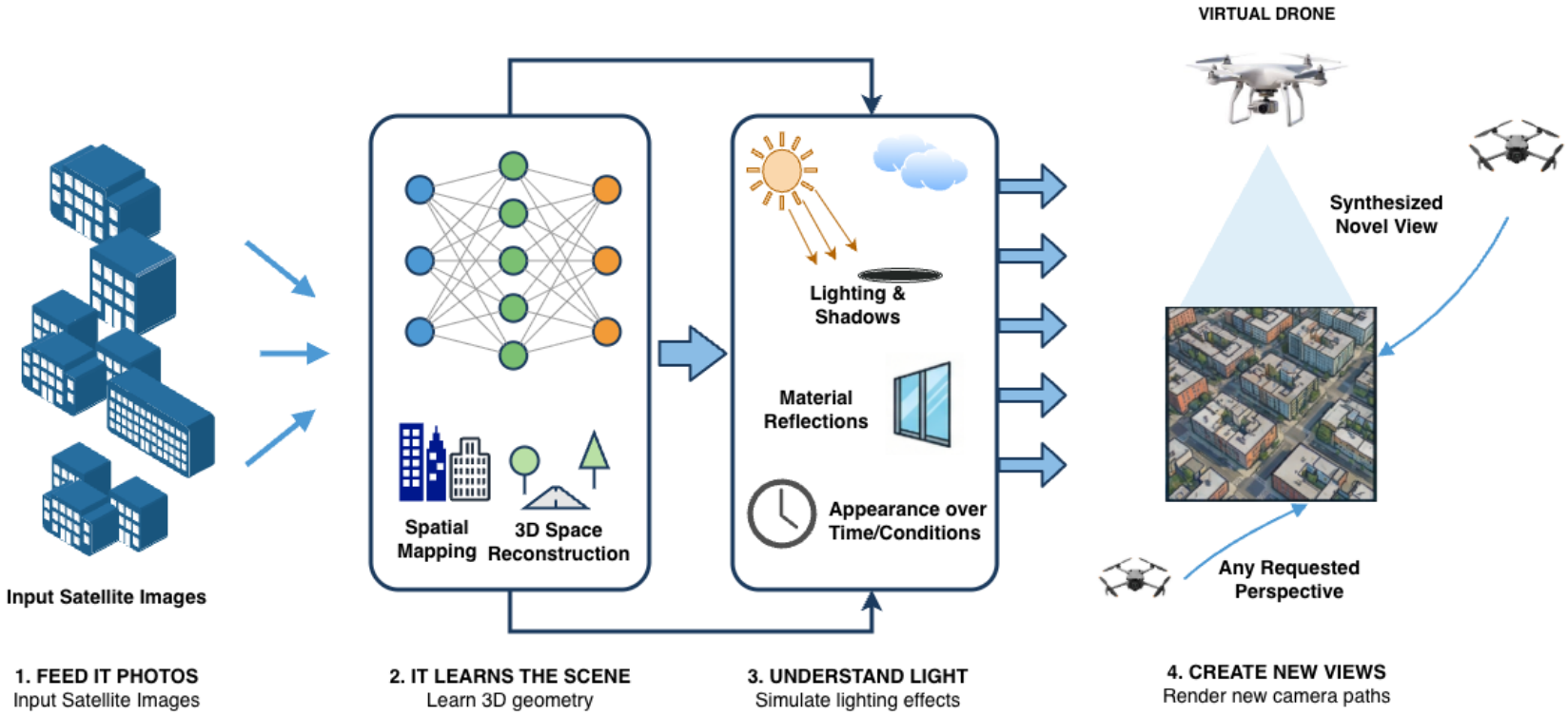
Neural Radiance Fields (NeRF)



Advantages & Drawbacks of NeRF

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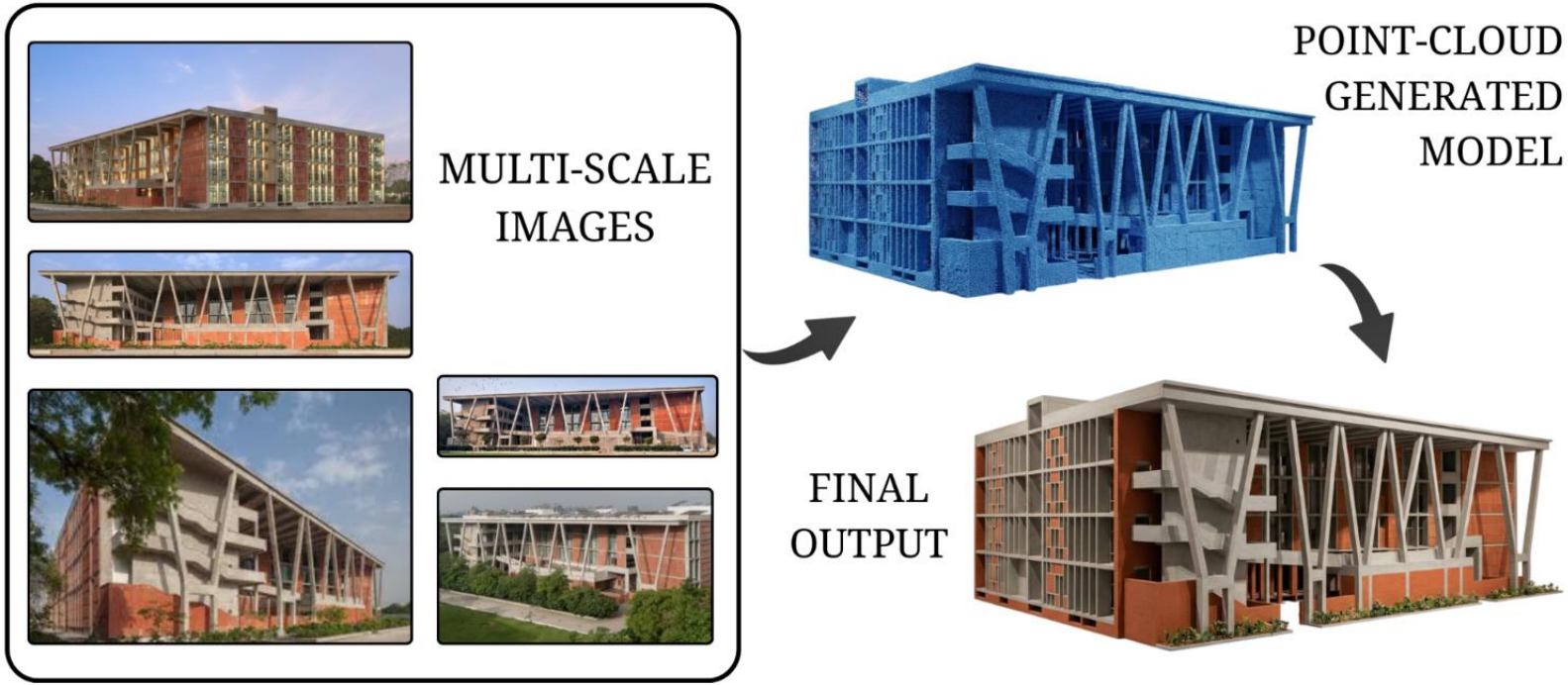
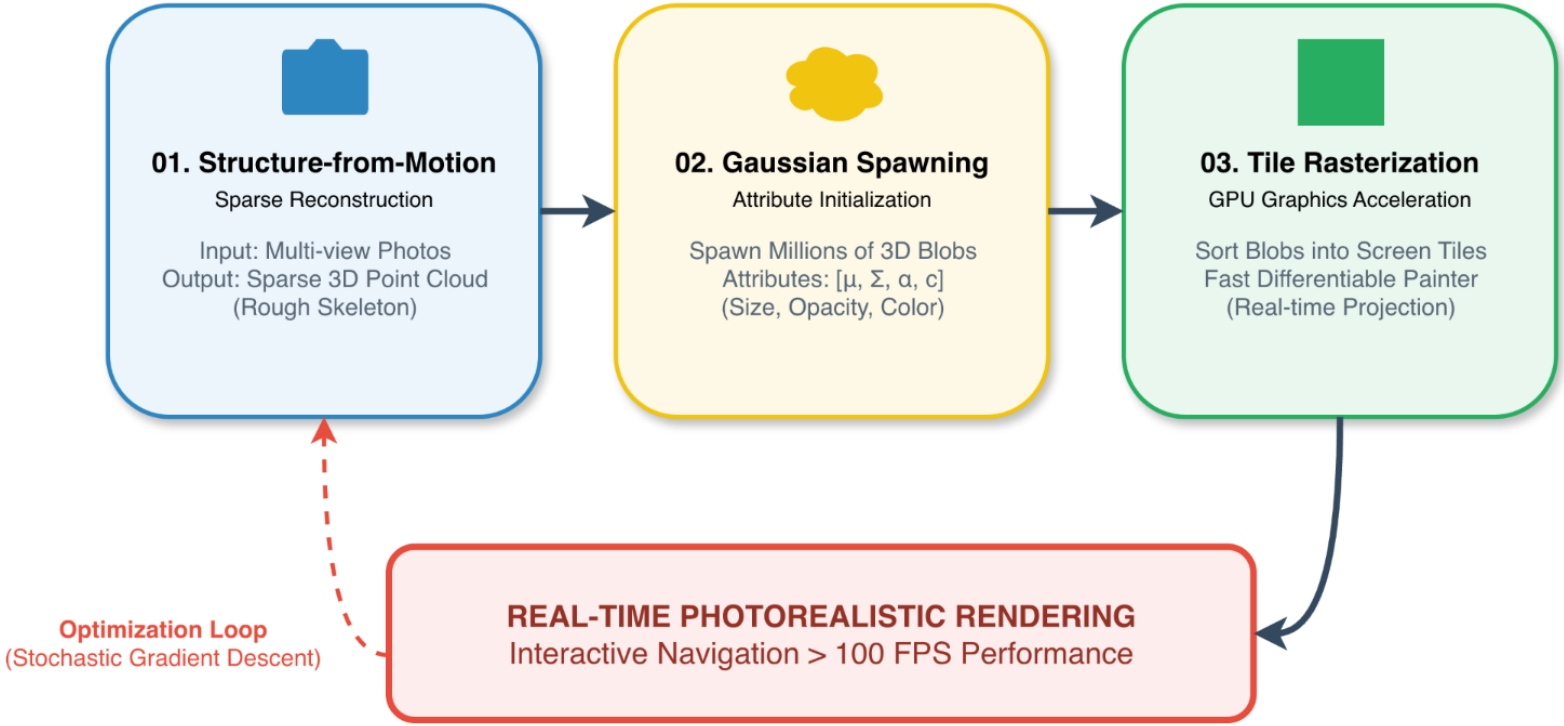
PROS	CONS
• Good at reflection and see through materials • Photorealistic view from new camera angles • Handles complex lighting and shadows • Better at 3D Detailing • Generates Ortho-Map • Can fill missing areas	• Very Slow in training • Cannot Render in real-time • Requires too many photos • Uses a Black-Box Approach • Struggles when data is from different Satellite types • Requires enormous computing power

3D Gaussian Splatting (3DGS)



3DGS in Nutshell



NeRF vs 3DGS

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	NeRF	3DGS
How it Works	**Neural Network** learns the scene	**Millions of 3D blobs** describe the scene
Training Speed	Hours to Days	Minutes to Few Hours (10x)
Rendering Speed	**Slow** - seconds per frame	**Real-time** - 100+ frames per second
Image Quality	Great but can be **blurry**	**Sharp Edges**, Higher Quality
Editability	Hard (it's a **black box**)	**Easy** to move, delete, or label blobs
Sparse Photos	**Struggles** with **few images**	Needs **extra tools**, but can handle it
City-Scale Use	Works with **chunking tricks**	Better suited for **parallel processing**

5 Major Research Areas Covered in This Survey

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Creating distortion-free, top-down maps from 3D satellite data like Google Maps, but hyper-accurate

Building detailed 3D models of entire cities by slicing them into smaller, manageable chunks.

Teaching AI to understand how the sun moves and how shadows change, so buildings are correctly shaped

Techniques to build accurate 3D models even when only a handful of satellite images are available

Automatically detects new construction, demolished buildings, or changes to the landscape across years

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Shadows, Seasons & Changing Skies

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The Problem: Shifting Sun

- **Same street** having **different appearance** (8 AM vs 4 PM)
- **Dynamic shadows** generate **distorted building shapes**
- **Seasonal variation** like green trees vs winter skeletons
- **Transient objects** like vehicles lead to ghost artifacts

The AI Solution

- **S-NeRF** uses **sun position** to estimate the building features per image
- **Shadow disentanglement** helps in true surface color extraction
- **SatGS** uses seasonal encoding
- **Dynamic filtering** considers moving objects as noise

Overcoming Top-Down Bias

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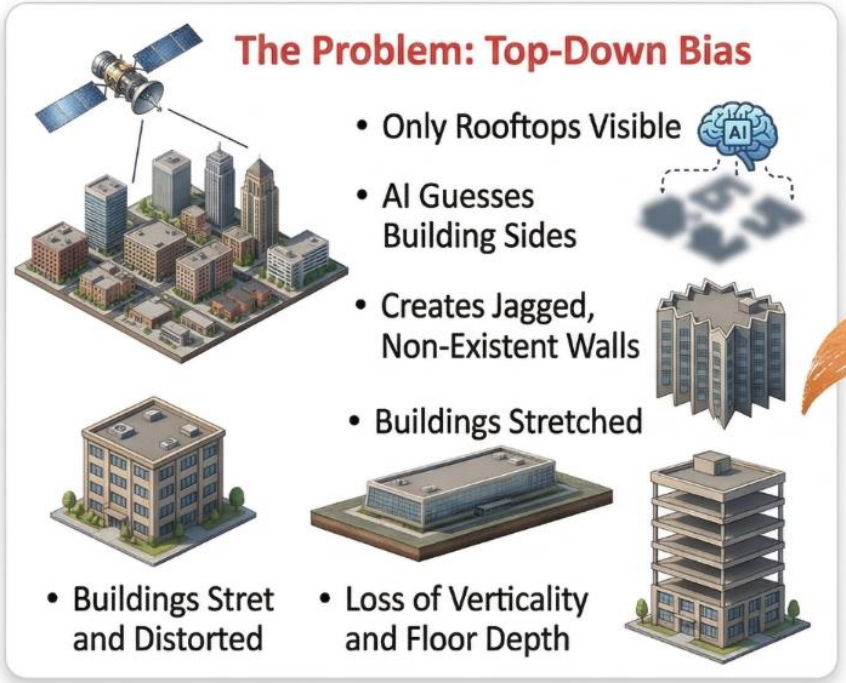

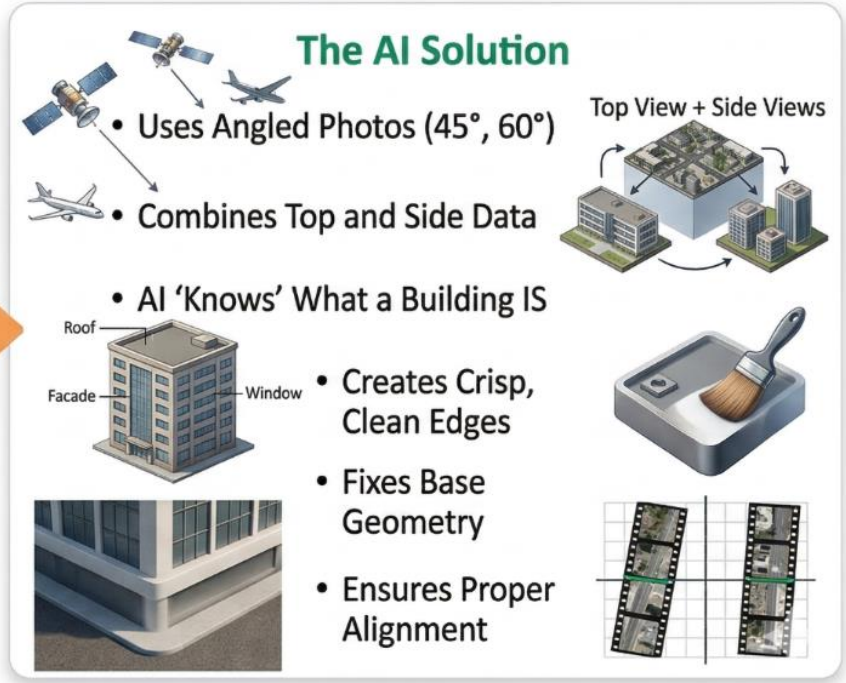

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Computational Overhead

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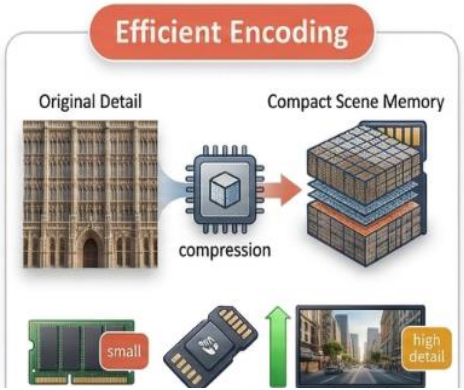
16+ Hrs	100+ FPS	10x+	Millions
NeRF training time	**3DGS Rendering**	**Speed advantage**	**Gaussian blobs**
for one large urban scene	*frame per second (realtime)*	*3DGS vs NeRF training*	*needed for single city*



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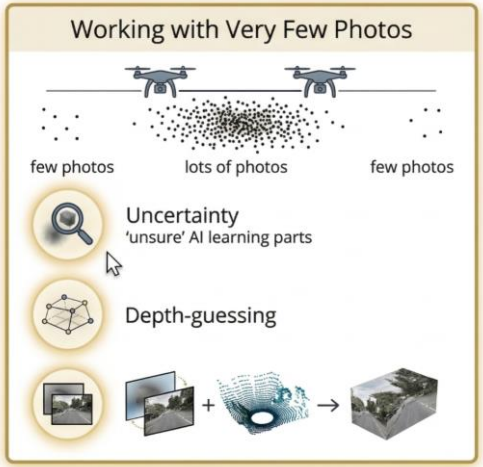
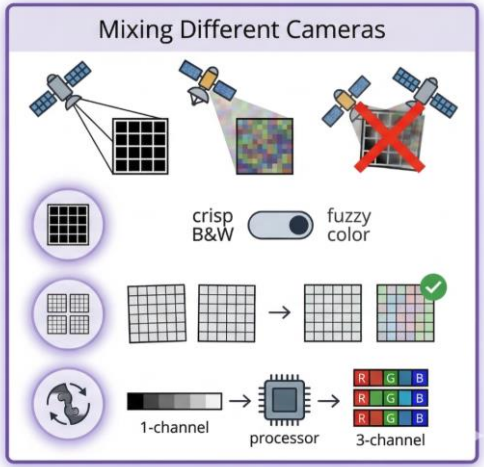
Overcoming Data Limits and Mixing Cameras



- **UPNeRF:** Prioritizes easy parts, flags uncertainty
- **FSGS:** Guesses depth from few photos
- **DS-NeRF:** Uses LiDAR for depth guidance

- **EOGS++:** Uses sharp black-and-white directly.
- **Internal Bundle Adjustment:** Automatically fixes camera misalignments.
- **3-PAN Strategy:** Converts single to 3-channel data.



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Results and Evaluation Matrix

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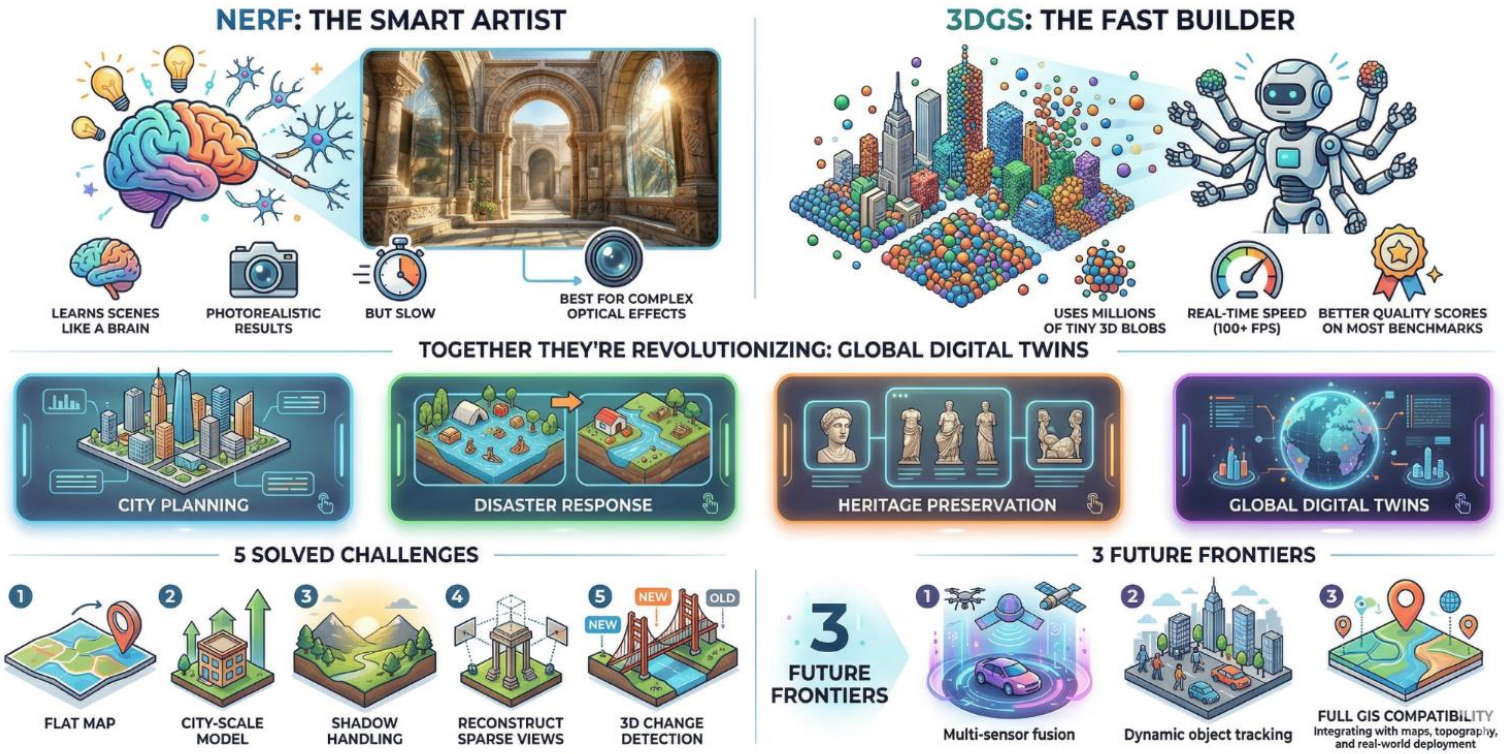
PSNR — **Peak Signal-to-Noise Ratio**
Measures sharpness (higher value means cleaner image)

SSIM — **Structural Similarity Index**
How similar the structure looks (1.0 = perfect)

LPIPS — **Learned Perceptual Similarity**
How realistic it looks to human eyes (lower the value, better it is)

Landmark	NeRF PSNR	3DGS PSNR	NeRF SSIM	3DGS SSIM	Improvement
Bank of China Tower	22.6 dB	**34.1 dB**	0.546	**0.980**	**51% sharper!**
Gardens by the Bay	24.3 dB	**35.6 dB**	0.730	**0.985**	**47% sharper!**
Tokyo Tower	21.1 dB	**32.6 dB**	0.639	**0.968**	**55% sharper!**

Key Takeaways



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